

Statistics Practical Questions with R Solutions

1) Descriptive Statistics

Question:

Obtain descriptive statistics for the sample of daily production in meters of 30 carpet looms using MS Excel and R.

R Code:

```
x <- c(
16.2,16.8,15.9,15.6,15.9,16.6,
15.8,16,16,15.7,15.9,15.6,
15.8,16.4,16.3,16,16.8,15.6,
15.8,15.2,16,16.2,15.4,16.9,
16.3,15.9,16.4,16.1,15.7,16.3
)
```

```
mean(x)
median(x)
sd(x)
var(x)
summary(x)
```

2) King Penguin Hypothesis Test

Question:

Suppose mean weight of King Penguins last year was 15.4 kg.

For a sample of 35 penguins, mean weight = 14.6 kg and population SD = 2.5 kg.

At 0.05 significance level, test whether mean differs from last year.

R Code:

```
mu <- 15.4
xbar <- 14.6
sigma <- 2.5
n <- 35
```

```
z <- (xbar-mu)/(sigma/sqrt(n))
z
```

```
pvalue <- 2*pnorm(z)
pvalue
```

3) Sampling Distribution Question

Question:

Mean = 8000, standard deviation = 3200, n = 64.

Find:

- (i) $P(\bar{x} > 9000)$
- (ii) $P(\bar{x} < 8500)$

R Code:

```
mu <- 8000
sigma <- 3200
n <- 64
```

```
se <- sigma/sqrt(n)
```

```
1-pnorm(9000,mean=mu,sd=se)
```

```
pnorm(8500,mean=mu,sd=se)
```

4) Normal Distribution

Question:

In a sample from normal distribution with mean 18 and SD 4.8:

- (i) $P(x \leq 16)$
- (ii) $P(x > 16)$
- (iii) $P(x \leq 20)$

R Code:

```
mu <- 18
sigma <- 4.8

pnorm(16, mean=mu, sd=sigma)

1-pnorm(16, mean=mu, sd=sigma)

pnorm(20, mean=mu, sd=sigma)
```

5) Binomial Distribution

Question:

For binomial distribution with $n=10$ and $p=0.45$ find:

- (a) $P(r=8)$
- (b) $P(r>4)$
- (c) $P(r \leq 6)$

R Code:

```
n <- 10
p <- 0.45

dbinom(8, n, p)

1-pbinom(4, n, p)

pbinom(6, n, p)
```

6) English Quiz Problem

Question:

There are 12 MCQ questions. Each has 5 options with only 1 correct answer.
Find probability of getting four or less correct answers by random guessing.

R Code:

```
n <- 12
p <- 1/5

pbinom(4, n, p)
```

7) Hitavada Press Problem

Question:

Average life hypothesized = 14500 hours.

Population SD = 2100 hours.

Sample size = 25 and sample mean = 13000 hours.

At 0.01 significance level, test whether average life is less than 14500 hours.

R Code:

```
mu <- 14500
sigma <- 2100
n <- 25
xbar <- 13000

z <- (xbar-mu)/(sigma/sqrt(n))
```

z

```
pvalue <- pnorm(z)
pvalue
```

8) LEGO Toys

Question:

Toy weights are normally distributed with mean 17.46 grams and variance 375.67.
Find probability that a toy weighs more than 19 grams.

R Code:

```
mu <- 17.46
sigma <- sqrt(375.67)

1-pnorm(19,mean=mu,sd=sigma)
```

9) One Sample t-Test

Question:

Given sample mean = 83, sample SD = 12.5, sample size = 22.
Test $H_0: \mu = 70$ against $H_1: \mu > 70$ at 0.025 significance level.

R Code:

```
xbar <- 83
s <- 12.5
n <- 22
mu <- 70

t <- (xbar-mu)/(s/sqrt(n))
t

1-pt(t,df=n-1)
```

10) Exam Scores

Question:

Test scores follow normal distribution with mean 72 and SD 15.2.
Find percentage scoring 84 or more.

R Code:

```
(1-pnorm(84,mean=72,sd=15.2))*100
```

11) Normal Random Variable

Question:

A normal random variable has mean 22 and variance 25. Find:

- (i) $P(16.2 < X < 27.5)$
- (ii) $P(X > 29)$
- (iii) $P(X < 17)$
- (iv) $P(X < 15 \text{ or } X > 25)$

R Code:

```
mu <- 22
sigma <- 5

pnorm(27.5,mu,sigma)-pnorm(16.2,mu,sigma)

1-pnorm(29,mu,sigma)
pnorm(17,mu,sigma)
```

```
pnorm(15,mu,sigma)+(1-pnorm(25,mu,sigma))
```

12) Poisson Bridge Problem

Question:

If 12 cars cross a bridge per minute on average,
find probability that 17 or more cars cross in a minute.

R Code:

```
1-ppois(16,12)
```

13) Bottle Filling Machine

Question:

Bottle contents are normally distributed with mean 298 ml and SD 3 ml.
Find probability that average content of 6 bottles is less than 295 ml.

R Code:

```
mu <- 298  
sigma <- 3  
n <- 6
```

```
se <- sigma/sqrt(n)
```

```
pnorm(295,mean=mu,sd=se)
```

14) Poisson Distribution Mean 7

Question:

Random variable X follows Poisson distribution with mean 7.

Find:

- (i) $P(X < 5)$
- (ii) $P(X > 10)$
- (iii) $P(4 < X < 16)$

R Code:

```
lambda <- 7
```

```
ppois(4,lambda)
```

```
1-ppois(10,lambda)
```

```
ppois(15,lambda)-ppois(4,lambda)
```

15) t-Test Problem

Question:

Given $\bar{x} = 94.3$, sample SD = 8.4, $n = 6$.

Test $H_0: \mu = 100$ against $H_1: \mu < 100$ at 0.05 significance level.

R Code:

```
xbar <- 94.3  
s <- 8.4  
n <- 6  
mu <- 100
```

```
t <- (xbar-mu)/(s/sqrt(n))  
t
```

```
pt(t,df=n-1)
```

16) Light Bulb Claim

Question:

Manufacturer claims mean lifetime > 10000 hours.
Sample of 30 bulbs gives mean = 9900 hours and SD = 120 hours.
At 0.05 significance level, test the claim.

R Code:

```
mu <- 10000
xbar <- 9900
sigma <- 120
n <- 30

z <- (xbar-mu)/(sigma/sqrt(n))
z
pnorm(z)
```

17) Coin Tosses

Question:

Find probabilities in 60 tosses of fair coin:
(i) heads occur exactly 20, 25 or 30 times
(ii) less than 20 times
(iii) between 20 and 30 times

R Code:

```
n <- 60
p <- 0.5

dbinom(20,n,p)+dbinom(25,n,p)+dbinom(30,n,p)

pbinom(19,n,p)

pbinom(30,n,p)-pbinom(19,n,p)
```

18) Binomial Successes

Question:

There are 10 independent trials with probability of success = 0.6.
Find probability of exactly 5 successes.

R Code:

```
dbinom(5,10,0.6)
```

19) Cookie Fat Label Test

Question:

Food label says saturated fat is at most 2 grams.
For 35 cookies, sample mean = 2.1 grams and sample SD = 0.3 grams.
At 0.05 significance level, test the claim.

R Code:

```
xbar <- 2.1
mu <- 2
s <- 0.3
n <- 35

t <- (xbar-mu)/(s/sqrt(n))
```

t

1-pt(t,df=n-1)

20) Employees Late Problem

Question:

Probability of employee being late = 0.4.

If 5 employees work in pharmacy, find probabilities that 0,1,2,3,4 or 5 employees are late simultaneously.

R Code:

n <- 5

p <- 0.4

dbinom(0,n,p)

dbinom(1,n,p)

dbinom(2,n,p)

dbinom(3,n,p)

dbinom(4,n,p)

dbinom(5,n,p)
